

SonarQube Testing of Java, JS, and Python Code (Two-Server Setup)

Goal

Set up two separate servers:

-  **Build Server:** for compiling code and running SonarQube analysis
-  **SonarQube Server:** for hosting SonarQube and viewing results

Test static analysis using SonarQube for Java, JavaScript, and Python projects.

1. SonarQube Server Setup

OS Info

```
uname -a
# Linux ip-172-31-20-239 ... x86_64 GNU/Linux
```

Install Prerequisites

```
sudo apt update -y
sudo apt install -y unzip
sudo apt install openjdk-17-jre-headless -y
```

Download and Extract SonarQube

```
wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-25.4.0.105899.zip
unzip sonarqube-25.4.0.105899.zip
```

Start SonarQube

```
cd sonarqube-25.4.0.105899/bin/linux-x86-64
./sonar.sh start
```

```
./sonar.sh status
```

✓ Expected Output:

```
Starting SonarQube...  
Started SonarQube.  
SonarQube is running (PID)
```

Open SonarQube in Browser

```
curl ifconfig.me  
# Output: 18.232.140.167
```

Access SonarQube at:

http://44.202.193.53:9000

Login: admin / admin Reset admin Password

2. Configure SonarQube Projects (Web UI)

For each language (Java / JS / Python):

Go to Projects → Create Project

The screenshot shows the SonarQube web interface. The top navigation bar includes 'Projects', 'Issues', 'Rules', 'Quality Profiles', 'Quality Gates', 'Administration', and 'More'. A yellow warning banner at the top states: 'Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)'. On the left, there are filters for 'Quality Gate' (Passed: 3, Failed: 0) and 'Security' (Info: 2, Low: 0, Medium: 0, High: 0, Blocker: 1). The main area displays a list of projects. The first project is 'java PUBLIC', which is 'Passed'. It shows 'Last analysis: 7 minutes ago - 1.5k Lines of Code - HTML, Java, ...' and metrics: Security (E 23), Reliability (D 5), Maintainability (A 15), Hotspots Reviewed (E 0.0%), Coverage (0.0%), and Duplications (4.4%). The second project is 'js PUBLIC', also 'Passed', with 'Last analysis: 2 minutes ago - 555 Lines of Code - TypeScript, JavaScript, ...' and metrics: Security (A 0), Reliability (C 1), Maintainability (A 7), Hotspots Reviewed (A —), Coverage (0.0%), and Duplications (11.8%). The third project is 'python PUBLIC', 'Passed', with 'Last analysis: 38 seconds ago - 178 Lines of Code - Python' and metrics: Security (A 0), Reliability (C 3), Maintainability (A 10), Hotspots Reviewed (A —), Coverage (0.0%), and Duplications (0.0%). The 'Create Project' button is highlighted with a red box and an arrow. The bottom status bar shows 'SonarQube™ technology is powered by SonarSource SA', 'Community Build - v25.4.0.105899 - HQR MODE', 'LGPL v3', 'Community', 'Documentation', 'Plugins', 'Web API', and a weather widget showing '35°C Mostly sunny'.

choose Local project

This screenshot shows the same SonarQube interface as the previous one, but with the 'Create Project' dropdown menu open. The 'Local project' option is highlighted with a red box and an arrow. The 'Import from DevOps Platforms' option is also visible below it. The rest of the page content remains the same.

give Required details

 Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#) 

1 of 2

Create a local project

Project display name * 

Project key * 

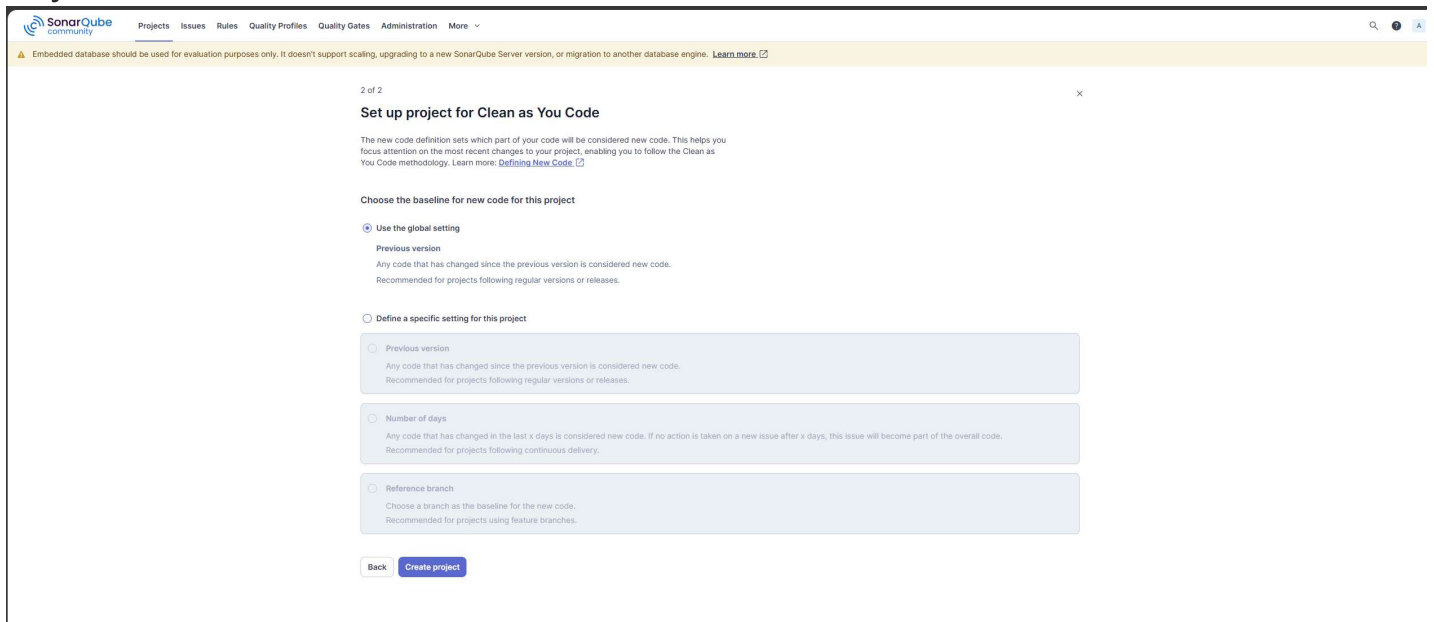
Main branch name *

The name of your project's default branch [Learn More](#) 

Select **Use global settings**, then choose **Locally**

Choose **Locally** Enter:

- Project key (e.g., `java` , `js` , or `python`)
- Project name



The screenshot shows the 'Set up project for Clean as You Code' configuration page in SonarQube. The page is titled '2 of 2' and includes a close button (X). The main heading is 'Set up project for Clean as You Code'. Below this, there is a paragraph explaining that the new code definition sets which part of the code will be considered new code, helping to focus attention on the most recent changes. A link to 'Defining New Code' is provided. The section 'Choose the baseline for new code for this project' has two radio button options: 'Use the global setting' (which is selected) and 'Define a specific setting for this project'. Under 'Define a specific setting for this project', there are three sub-options: 'Previous version', 'Number of days', and 'Reference branch', each with a brief description and a recommended use case. At the bottom, there are 'Back' and 'Create project' buttons.

SonarQube community

Projects Issues Rules Quality Profiles Quality Gates Administration More

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java-1 / main

Overview Issues Security Hotspots Measures Code Activity

Project Settings Project Information

Analysis Method

Use this page to manage and set-up the way your analyses are performed.

How do you want to analyze your repository?

With Jenkins

With GitHub Actions

With Bitbucket Pipelines

With GitLab CI

With Azure Pipelines

Other CI
SonarQube Community Build integrates with your workflow no matter which CI tool you're using.

Locally
Use this for testing or advanced use-case. Other modes are recommended to help you set up your CI environment.

Generate a token:

- Click **Generate** → **Continue**

SonarQube community

Projects Issues Rules Quality Profiles Quality Gates Administration More

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java-1 / main

Overview Issues Security Hotspots Measures Code Activity

Project Settings Project Information

Analysis Method / Locally

Analyze your project

We initialized your project on SonarQube Community Build, now it's up to you to launch analyses!

1 Provide a token

Generate a project token

Use existing token

Token name *

Analyze "java-1"

Expires in

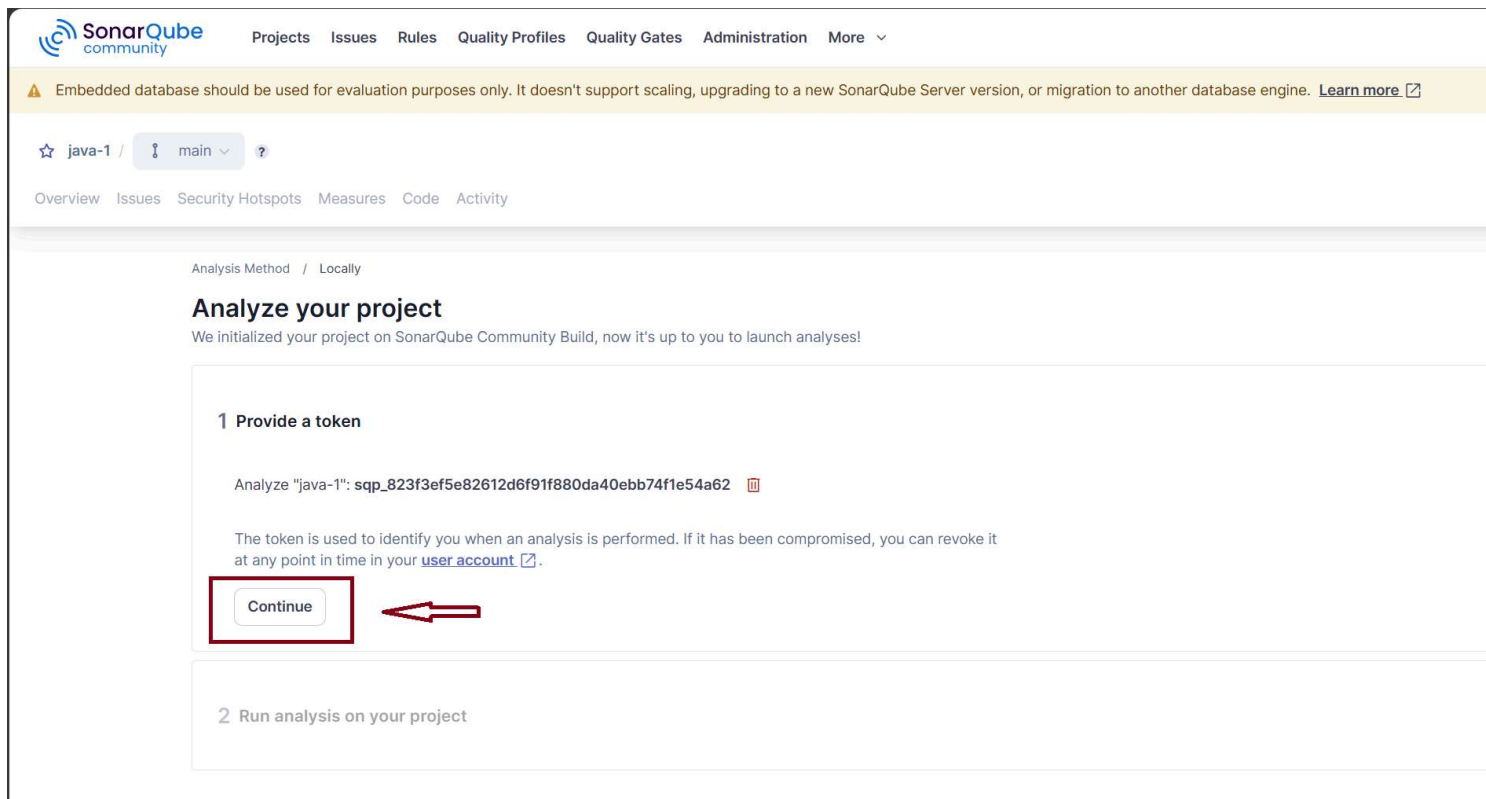
30 days

Generate

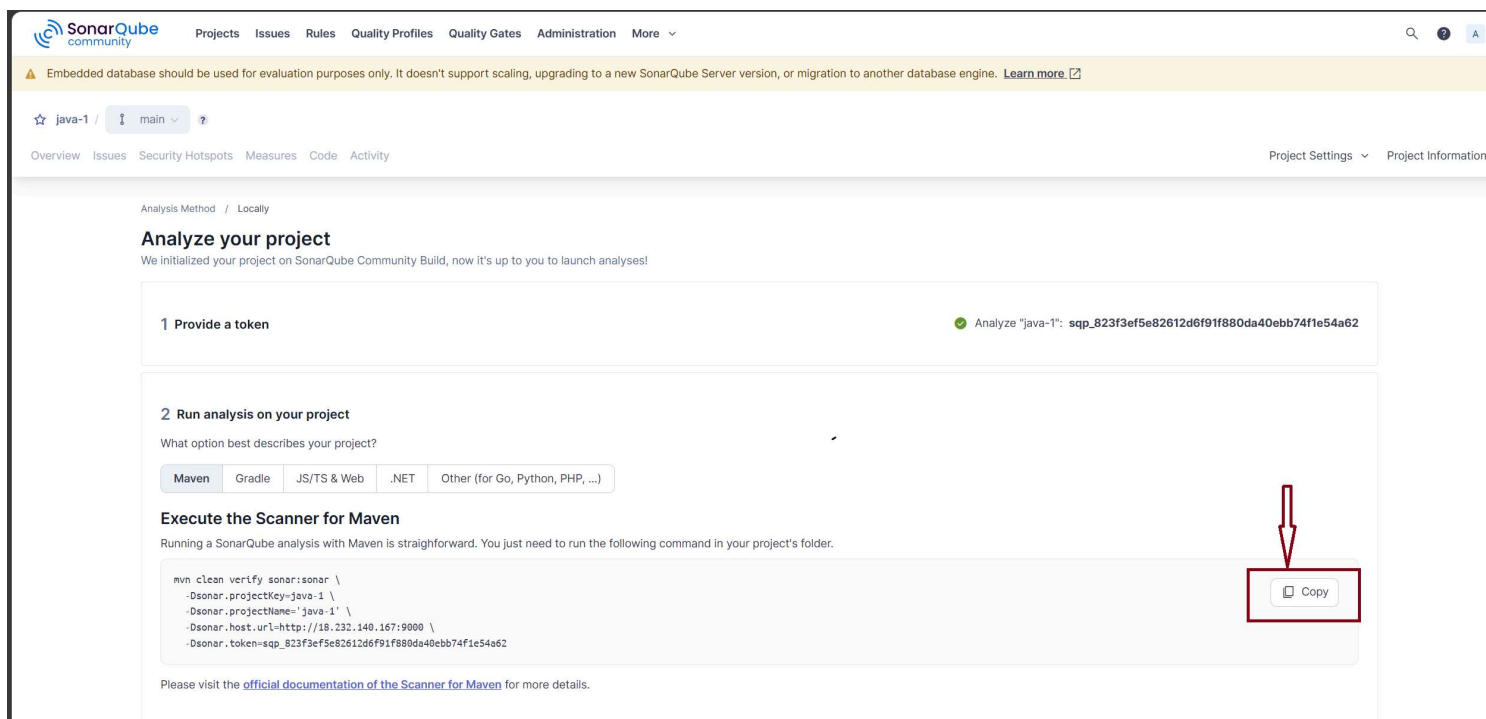
Please note that this token will only allow you to analyze the current project. If you want to use the same token to analyze multiple projects, you need to generate a global token in your [user account](#). See the [documentation](#) for more information.

The token is used to identify you when an analysis is performed. If it has been compromised, you can revoke it at any point in time in your [user account](#).

2 Run analysis on your project



In 2. choose the package/code and copy the code



3. Build Server Setup

Install General Prerequisites

```
apt update -y
apt install git -y
apt install openjdk-17-jre-headless -y
apt install maven -y
apt install npm -y
```

Java Project Analysis (Maven)

Clone and Build Java Project

```
git clone https://github.com/<your-java-repo>.git
cd Java-Blog
mvn package
```

```
root@ip-172-31-26-213:~/Documentation/java_code# ll
total 64
drwxr-xr-x 4 root root 4096 Apr 13 09:33 ./
drwxr-xr-x 6 root root 4096 Apr 13 09:33 ../
-rw-r--r-- 1 root root 15759 Apr 13 09:33 MySQLDesign.mwb
-rw-r--r-- 1 root root 15550 Apr 13 09:33 MySQLDesign.mwb.bak
-rw-r--r-- 1 root root 4494 Apr 13 09:33 pom.xml
-rw-r--r-- 1 root root 7262 Apr 13 09:33 readme.md
drwxr-xr-x 3 root root 4096 Apr 13 09:33 src/
drwxr-xr-x 7 root root 4096 Apr 13 09:33 target/
```

Run SonarQube Analysis

```
mvn clean verify sonar:sonar \
  -Dsonar.projectKey=java \
  -Dsonar.projectName='java' \
  -Dsonar.host.url=http://18.232.140.167:9000 \
  -Dsonar.token=sqp_XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

✅ Output will show analysis and upload to SonarQube.

JavaScript Project Analysis (NPM + SonarScanner)



Clone JS Project and Setup

```
git clone https://github.com/<your-js-repo>.git
cd js_code/
npm install
npm install -g @sonar/scan
```

```
root@ip-172-31-26-213:~/Documentation/js_code# ll
total 208
drwxr-xr-x 4 root root  4096 Apr 13 09:33 ./
drwxr-xr-x 6 root root  4096 Apr 13 09:33 ../
drwxr-xr-x 3 root root  4096 Apr 13 09:33 dist/
-rw-r--r-- 1 root root 172684 Apr 13 09:33 package-lock.json
-rw-r--r-- 1 root root  1086 Apr 13 09:33 package.json
-rw-r--r-- 1 root root  6041 Apr 13 09:33 readme.md
drwxr-xr-x 4 root root  4096 Apr 13 09:33 src/
-rw-r--r-- 1 root root   322 Apr 13 09:33 tsconfig.json
-rw-r--r-- 1 root root  1317 Apr 13 09:33 webpack.config.js
```

The screenshot shows the SonarQube Community Build web interface. The top navigation bar includes links for Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. A yellow warning banner at the top states: "Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)".

The main content area is titled "Analyze your project" and includes the subtext: "We initialized your project on SonarQube Community Build, now it's up to you to launch analyses!".

1 Provide a token

A green checkmark indicates success: "Analyze 'js-1': sqp_cceda467acaf4cce33d1655a97752a8dc7d10944".

2 Run analysis on your project

What option best describes your project?

Buttons: Maven, Gradle, **JS/TS & Web**, .NET, Other (for Go, Python, PHP, ...)

Install the Scanner for npm projects

npm install -g @sonar/scan Copy

Please visit the [official documentation of the Scanner for npm projects](#) for more details.

Execute the Scanner

Running a SonarQube analysis with the Scanner for npm projects is straightforward. You just need to run the following command in your project's folder.

```
sonar \
  -Dsonar.host.url=http://18.232.149.167:9090 \
  -Dsonar.token=sqp_cceda467acaf4cce33d1655a97752a8dc7d10944
```

Copy

Please visit the [official documentation of the Scanner for npm projects](#) for more details.



Create sonar-project.properties

```
vi sonar-project.properties
```

Paste the following:


```
sonar.projectKey=js
sonar.projectName=js
sonar.projectVersion=1.0
sonar.sources=.
sonar.host.url=http://18.232.140.167:9000
sonar.login=sqp_ff83de42e0c3931f39f99357a83f4c0fa98efbc6
```

Run SonarQube Scanner

```
sonar-scanner
```

✓ Results will be uploaded to the SonarQube server.

Python Project Analysis

Prepare and Analyze Python Project

```
cd python_code/
```

```
root@ip-172-31-26-213:~/Documentation/python_code# ll
total 20
drwxr-xr-x 3 root root 4096 Apr 13 09:33 ./
drwxr-xr-x 6 root root 4096 Apr 13 09:33 ../
drwxr-xr-x 4 root root 4096 Apr 13 09:33 app/
-rw-r--r-- 1 root root 6514 Apr 13 09:33 readme.md
```

The screenshot shows the SonarQube community web interface. At the top, there's a navigation bar with links like Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. A yellow warning banner at the top states: "Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)".

The main content area is titled "1 Provide a token" and "2 Run analysis on your project". Under "2 Run analysis on your project", there are buttons for "Maven", "Gradle", "JS/TS & Web", ".NET", and "Other (for Go, Python, PHP, ...)". Below these, there are buttons for "Linux", "Windows", and "macOS".

The "Download and unzip the Scanner for Linux" section provides instructions to visit the [official documentation of the Scanner](#) to download the latest version and add the `bin` directory to the `PATH` environment variable. It also includes a section "Execute the Scanner" with the command: `sonar-scanner \`
`-Dsonar.projectKey=js-1 \`
`-Dsonar.sources=. \`
`-Dsonar.host.url=http://18.232.140.167:9000 \`
`-Dsonar.token=sqp_cceda467acaf4cce33d1655a97752a8dc7d10944`

A "Copy" button is next to the command. At the bottom, it says "Please visit the [official documentation of the Scanner](#) for more details."

Run SonarQube Scanner

```
sonar-scanner \  
-Dsonar.projectKey=python \  
-Dsonar.sources=. \  
-Dsonar.host.url=http://44.202.193.53:9000 \  
-Dsonar.token=sqp_e36b7c7b5af01a866631fd7a9dff62cf5b0c3c59
```

```
[INFO] ScannerEngine: Analysis report compressed in 1475ms, zip size=11.1 MB  
[INFO] ScannerEngine: Analysis report uploaded in 160ms  
[INFO] ScannerEngine: ANALYSIS SUCCESSFUL, you can find the results at: http://18.232.140.167:90  
[INFO] ScannerEngine: Note that you will be able to access the updated dashboard once the server  
[INFO] ScannerEngine: More about the report processing at http://18.232.140.167:9000/api/ce/task  
[INFO] ScannerEngine: Analysis total time: 1:19.266 s  
[INFO] ScannerEngine: SonarScanner Engine completed successfully
```

 Python code analysis will appear in the SonarQube dashboard.

The screenshot shows the SonarQube Community web interface. At the top, there's a navigation bar with links like Projects, Issues, Rules, Quality Profiles, Quality Gates, Administration, and More. Below this, a banner message states: "Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)".

The main content area shows the project dashboard for the 'main' branch. It includes a 'Quality Gate' section with a green checkmark and the word 'Passed'. Below this, a warning message says: "The last analysis has warnings. [See details](#)".

The dashboard also features a 'New Code' and 'Overall Code' tab. Under 'Overall Code', there are several metrics displayed in a grid:

- Security:** 16 Open issues (D)
- Reliability:** 39 Open issues (E)
- Maintainability:** 2.8k Open issues (A)
- Accepted issues:** 0 (Valid issues that were not fixed)
- Coverage:** 0.0% (On 71k lines to cover)
- Duplications:** 18.6% (On 207k lines)
- Security Hotspots:** 91 (E)

4. View Results in SonarQube

Open browser → <http://18.232.140.167:9000> → Select your project dashboard

✓ Review for each project:

- Code smells
- Bugs
- Vulnerabilities
- Test coverage (if applicable)

Summary

Language	Build Tool	Analysis Command
Java	Maven	<code>mvn clean verify sonar:sonar ...</code>
JS	NPM + Scanner	<code>sonar-scanner</code> with <code>sonar-project.properties</code>
Python	Sonar Scanner	<code>sonar-scanner -Dsonar.projectKey=...</code>

 All results are published to: <http://44.202.193.53:9000>

InPrivate

Projects - SonarQube Community X py_code - Overview - SonarQube X +

← ↻ ⚠ Not secure | 44.202.193.53:9000/projects 🔍 ☆ ⋮

SonarQube community

Projects Issues Rules Quality Profiles Quality Gates Administration More ▾

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

My Favorites All

Filters

Quality Gate

- Passed 3
- Failed 0

Security

- A ≥ 0 info issues 1
- B ≥ 1 low issue 0
- C ≥ 1 medium issue 0
- D ≥ 1 high issue 1
- E ≥ 1 blocker issue 1

Reliability

- A ≥ 0 info issues 0
- B ≥ 1 low issue 0
- C ≥ 1 medium issue 1
- D ≥ 1 high issue 1
- E ≥ 1 blocker issue 1

Create Project ▾

Search for projects... Perspective Overall Status ▾ Sort by Name ▾ 3 project(s)

☆ java-1 PUBLIC Passed

Last analysis: 16 minutes ago · 3.4k Lines of Code · Java, JSP, ...

A 0

Security

D 73

Reliability

A 162

Maintainability

E 0.0%

Hotspots Reviewed

0.0%

Coverage

5.2%

Duplications

☆ js-1 PUBLIC Passed

Last analysis: 4 minutes ago · 1k Lines of Code · TypeScript, HTML, ...

E 1

Security

C 38

Reliability

A 26

Maintainability

—

Hotspots Reviewed

0.0%

Coverage

20.6%

Duplications

☆ py_code PUBLIC Passed

Last analysis: 2 minutes ago · 144k Lines of Code · Python, XML

D 16

Security

E 39

Reliability

A 2.8k

Maintainability

E 0.0%

Hotspots Reviewed

0.0%

Coverage

18.6%

Duplications

SonarQube™ technology is powered by SonarSource SA Community Build · v25.4.0.105899 · HQR MODE LGPL v3 Community Documentation Plugins Web API